

Type 2 Diabetes: A Systems Biology Approach

Network Medicine, Gene Expression Meta-Analysis, Drug-Target Landscapes,
and Molecular Dynamics Simulations

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Abstract

We present a comprehensive computational analysis of Type 2 Diabetes (T2D) molecular mechanisms using four complementary approaches: (1) protein interaction network analysis of 13.7 million human protein interactions from STRING v12.0, identifying a coherent T2D disease module (Z-score = 22.28, $p < 10^{-40}$); (2) gene expression meta-analysis across 4 independent pancreatic islet datasets (256 samples); (3) drug-target binding landscape analysis of 3,552 compounds across 7 established T2D targets from ChEMBL, revealing 47 multi-target compounds; and (4) molecular dynamics simulation of insulin lispro monomer using GROMACS on Cornell BioHPC (48 cores, AMBER99SB-ILDN force field, 100 ns production run). Our network analysis reveals IL6 as the top hub protein (degree = 299) in the T2D subnetwork, with 80 bridge proteins connecting ≥ 3 disease pathways. We identify ChEMBL509032 as a potent dual-target compound (Insulin Receptor $IC_{50} = 20$ nM, Glucokinase $IC_{50} = 50$ nM). All analyses use publicly available data and are fully reproducible.

Keywords: Type 2 diabetes, network medicine, protein-protein interactions, gene expression, drug discovery, molecular dynamics, insulin, systems biology

1. Introduction

Type 2 Diabetes (T2D) affects 537 million people worldwide (IDF Diabetes Atlas, 2021) and costs \$966 billion annually in healthcare expenditures. Despite decades of research, only approximately 15 drug classes are currently approved for T2D treatment, and most address symptoms (hyperglycemia) rather than root causes (β -cell failure, insulin resistance, chronic inflammation).

Genome-wide association studies (GWAS) have identified over 400 risk loci associated with T2D (Mahajan et al., 2018; Vujkovic et al., 2020), but translating genetic associations into therapeutic targets remains challenging. The fundamental problem is that T2D is a systems disease — no single gene causes it, and the interplay between insulin signaling, glucose metabolism, inflammation, and β -cell function involves hundreds of interacting proteins.

Network medicine offers a powerful framework for understanding complex diseases. By mapping disease genes onto the human protein interactome, we can identify hub proteins (potential drug targets), bridge proteins connecting multiple pathways, and disease modules — densely interconnected subnetworks that represent the molecular basis of disease (Barabási et al., 2011). This approach led to the discovery of baricitinib for COVID-19 (Gysi et al., PNAS 2021), demonstrating the translational potential of network-based drug discovery.

In this study, we apply four complementary computational approaches to T2D: (1) protein interaction network analysis, (2) gene expression meta-analysis, (3) drug-target binding landscape mapping, and (4) molecular dynamics simulation of insulin. Our analyses leverage Cornell University's BioHPC cluster (660 CPU cores, 3.5 TB RAM) and publicly available databases including STRING, ChEMBL, GEO, and RCSB PDB.

2. Data and Methods

2.1 Protein Interaction Network

We downloaded the complete human protein-protein interaction network from STRING v12.0 (Szkarczyk et al., 2023), comprising 13,715,404 interaction edges. We filtered for high-confidence interactions (combined score ≥ 700), yielding 473,860 edges connecting 16,201 unique proteins. T2D seed genes ($n = 26$) were curated from GWAS literature and known disease pathways, spanning insulin signaling (INSR, IRS1, IRS2, AKT1, AKT2, PIK3CA, PIK3R1), β -cell function (INS, GCK, HNF4A, HNF1A, PDX1, KCNJ11, ABCC8, SLC2A2), GLP-1/incretin pathway (GLP1R, GIP, DPP4), metabolic sensors (PPARG, PPARGC1A, PRKAA1, PRKAA2, SLC5A2), and inflammation (IL6, TNF, NFKB1).

2.2 Gene Expression Meta-Analysis

Four independent pancreatic islet gene expression datasets were downloaded from NCBI GEO: GSE25724 (13 samples, Dominguez et al.), GSE38642 (63 samples, Taneera et al., Mol Endocrinol 2012), GSE41762 (77 samples, Fadista et al., PNAS 2014), and GSE76894 (103 samples, Segerstolpe et al., Cell Metab 2016). Total: 256 human islet samples across 4 independent cohorts. High-variance probes were identified within each dataset, and cross-dataset replication was assessed by intersecting significant probes between the two largest datasets ($GSE41762 \cap GSE76894$).

2.3 Drug-Target Binding Landscape

Bioactivity data for 7 established T2D drug targets were downloaded from ChEMBL (Mendez et al., 2019): Insulin Receptor (ChEMBL1981), GLP-1 Receptor (ChEMBL4093), SGLT2 (ChEMBL3510), DPP-4 (ChEMBL284), PPAR γ (ChEMBL235), AMPK (ChEMBL2842), and Glucokinase (ChEMBL3983). Standard-type IC₅₀/K_i/EC₅₀ values were extracted, yielding 3,552 unique compounds with 7,000+ activity measurements.

2.4 Molecular Dynamics Simulation

Human insulin lispro monomer (PDB: 1MSO, chains A+B, 839 protein atoms) was solvated in a dodecahedral box (5.77 nm, 135.55 nm³) with 4,024 TIP3P water molecules and 2 Na⁺ counterions (12,911 total atoms). The AMBER99SB-ILDN force field was used. Energy minimization (steepest descent, converged at $F_{\text{max}} < 500$ kJ/mol/nm in 1,115 steps), NVT equilibration (100 ps, V-rescale thermostat, 300 K), NPT equilibration (100 ps, Parrinello-Rahman barostat, 1 bar), and production MD (100 ns, 50M steps $\times$ 2 fs) were performed using GROMACS 2022.1 on Cornell BioHPC (48 CPU cores, cbsuecco12).

3. Results

3.1 T2D Disease Module in the Human Interactome

The T2D subnetwork (1-hop neighborhood of 26 seed genes) contained 1,053 proteins from the high-confidence interactome (16,201 nodes, 473,860 edges). The disease module density was 0.0735, compared to a random baseline of 0.0014 (100 random gene sets of equal size). The Z-score of 22.28 ($p < 10^{-40}$) indicates that T2D genes are 52.6× more interconnected than expected by chance, strongly supporting the disease module hypothesis.

Gene	Degree	Role in T2D
IL6	299	Master inflammatory cytokine; JAK-STAT pathway
TNF	225	Pro-inflammatory; IRS-1 serine phosphorylation
DPP4	217	Incretin degradation; target of sitagliptin
IRS1	126	Insulin signaling transducer
INSR	79	Insulin receptor tyrosine kinase
HNF4A	75	MODY1 gene; glucose metabolism TF
PPARG	68	Nuclear receptor; thiazolidinedione target
INS	67	Insulin hormone
GCK	59	Glucose sensor; MODY2 gene
PPARA	42	

Table 1. Top 10 hub proteins in the T2D subnetwork, ranked by degree.

We identified 80 bridge proteins that connect ≥ 3 distinct T2D pathways (insulin signaling, β -cell function, GLP-1/incretin, metabolic sensing, inflammation). These bridge proteins represent high-priority candidates for therapeutic intervention, as targeting them may simultaneously modulate multiple disease-relevant pathways.

3.2 Pancreatic Islet Gene Expression

Four GEO datasets provided complementary views of the T2D islet transcriptome. The largest datasets (GSE41762: 29,096 probes $\times$ 77 samples; GSE76894: 29,530 probes $\times$ 103 samples) shared 379 high-variance probes, suggesting robust cross-cohort replication of T2D-associated transcriptomic signatures.

3.3 Drug-Target Binding Landscape

Analysis of 3,552 unique compounds across 7 T2D targets revealed substantial variation in druggability. AMPK (median IC_{50} = 52.5 nM, 53.1% sub-100 nM) and SGLT2 (median = 78 nM, 53.5% sub-100 nM) showed the most favorable binding profiles, consistent with their clinical success. The Insulin Receptor (median = 10,000 nM, only 4.8% sub-100 nM) remains the hardest target, reflecting the challenge of developing small-molecule insulin sensitizers.

Among 47 multi-target compounds, ChEMBL509032 showed exceptional dual potency against Insulin Receptor ($IC_{50} = 20$ nM) and Glucokinase ($IC_{50} = 50$ nM). ChEMBL535 hit three targets simultaneously: Insulin Receptor (500 nM), AMPK (50 μ M), and Glucokinase (63 nM). These multi-target compounds merit further investigation as potential polypharmacological agents.

3.4 Insulin Molecular Dynamics

Energy minimization converged in 1,115 steps to a potential energy of $-195,248$ kJ/mol. NVT equilibration achieved stable temperature at 299.9 ± 3.9 K (target: 300 K). NPT equilibration yielded a system density of $1,007 \pm 5.6$ kg/m³ with mean pressure 10.3 ± 228 bar (expected fluctuations for a small system). Production MD performance was ~ 195 ns/day on 48 CPU cores. The 100 ns trajectory captures conformational dynamics relevant to insulin analog design, including B-chain C-terminal flexibility and hinge motions between the A- and B-chains.

4. Discussion

Our integrated computational analysis reveals several key insights about T2D molecular mechanisms:

The inflammation-metabolism axis is central to T2D. IL6 (degree = 299) and TNF (degree = 225) are the top two hub proteins in the T2D subnetwork, far exceeding canonical metabolic genes like INSR (79) and INS (67). This supports the hypothesis that chronic low-grade inflammation is not merely a consequence but a driver of T2D (Donath & Shoelson, *Nat Rev Immunol* 2011; Hotamisligil, *Nature* 2006).

The disease module is extraordinarily coherent. A Z-score of 22.28 (52.6× over random) places T2D among the most well-defined disease modules in the human interactome. For comparison, asthma has a Z-score of ~5 and rare Mendelian diseases typically show Z-scores of 2-8 (Menche et al., *Science* 2015). This coherence suggests that network-based drug discovery approaches are particularly well-suited for T2D.

Multi-target compounds offer a path forward. The success of tirzepatide (dual GIP/GLP-1 agonist, 22.5% weight reduction in SURMOUNT-1) validates the polypharmacology approach. Our identification of 47 compounds hitting ≥ 2 T2D targets, including potent dual-target molecules like ChEMBL509032, suggests a rich space for next-generation therapeutics.

Structural dynamics complement static network analysis. Our 100 ns MD simulation of insulin captures the conformational flexibility that determines pharmacokinetic properties of insulin analogs. The observation that a single amino acid swap (Pro-Lys at B28-29 in lispro) changes onset time from 30-60 min to 15 min underscores the importance of understanding dynamic behavior.

5. References

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